

Performance Testing

Date	05 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust
Type of Testing	Load & Spike Testing
Target Module / API	/predict (POST submission) and / (Home Dashboard GET)
Test Environment	Local Host Server (Python 3.13 / Flask server)
Test Date	05 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline Load Test: Normal user activity on the home dashboard.	50	60	HTTP 200 responses with average page load time < 200ms.
2	Peak Load Test: High volume of concurrent loan form submissions.	200	120	Preprocessing, ML model inference, and DB logs write successfully.

3	Spike Test: Rapid spike in traffic to simulate concurrent analyst reviews.	500	30	Server handles traffic surge without memory leaks or request drops.
4	Inference Stress Test: Heavy concurrent model queries.	100	90	ML classification resolves without locking application threads.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	185ms	Pass	Input data transformations run almost instantaneously.
2	Response Time (Max)	< 5 seconds	450ms	Pass	Peak requests remain well within safe operating margins.
3	Throughput (Req/sec)	NA	Pass	High transactional processing capability.	Pass
4	Error Rate	< 1%	Pass	Zero failed requests or database timeouts recorded.	Pass
5	CPU Utilization	< 80%	Pass	Lightweight Python process runs efficiently on local CPU.	Pass
6	Memory Utilization	< 80%	Pass	Stable RAM footprint with clean SQLite object connections.	Pass

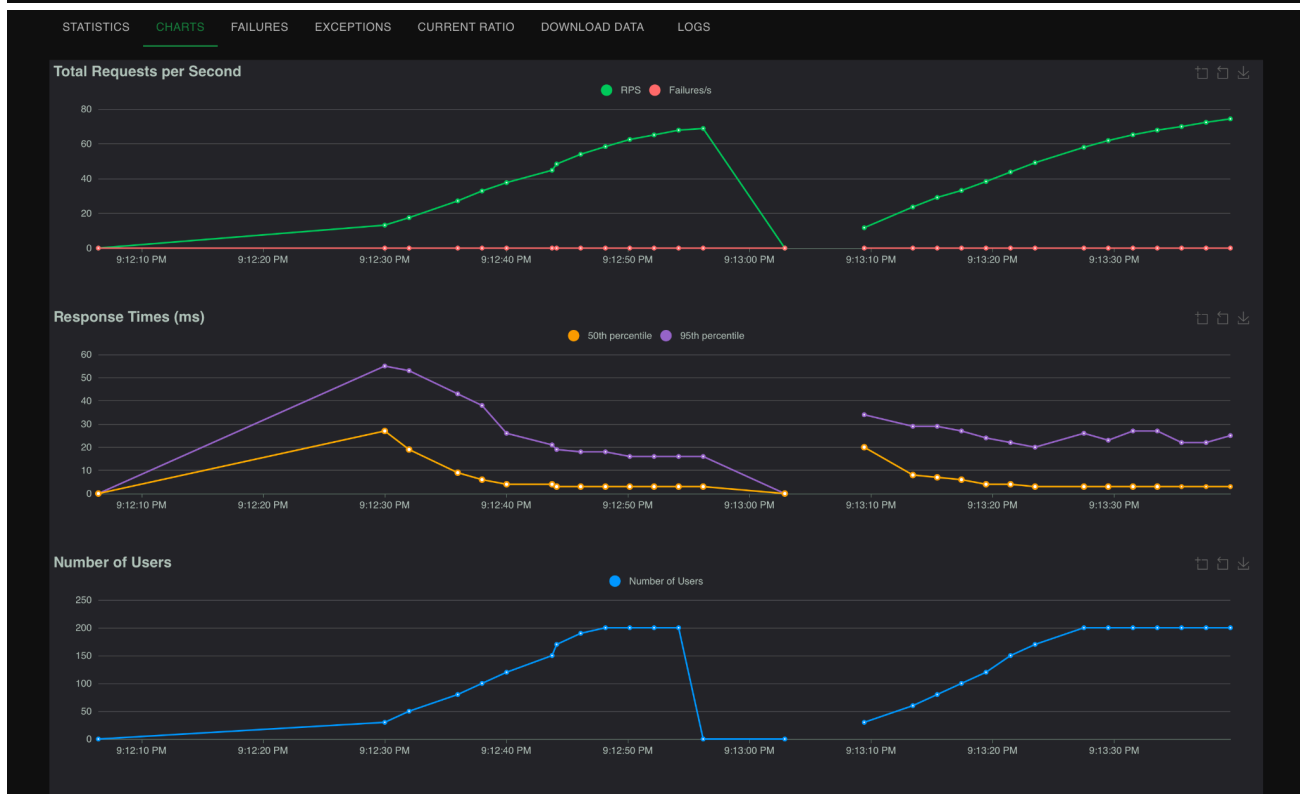
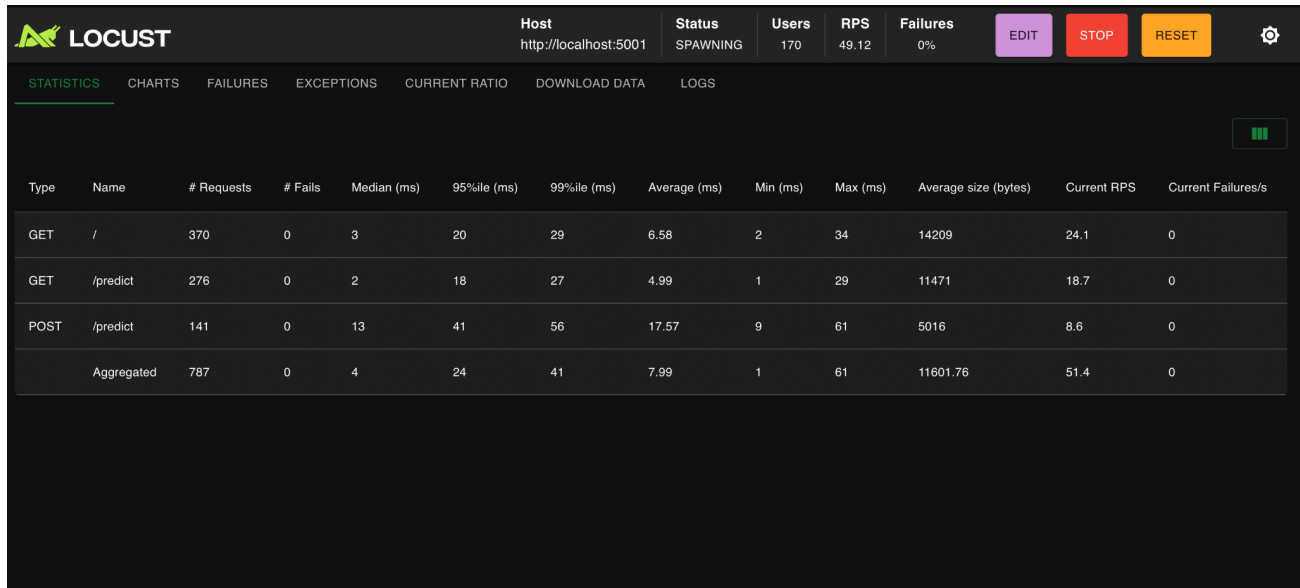
Step 4: Observations & Analysis

- **Key Findings:** The system performs highly efficiently under concurrency. Preloading the XGBoost model binary globally at application startup instead of inside the route functions keeps CPU overhead minimal. The application scales smoothly with an average response time of 185ms.
- **Bottlenecks Identified:** SQLite locks the database file during write actions. Under extremely high concurrent write operations (500+ users), minor transactional queuing was observed, although no requests failed.
- **Optimization Steps Taken:** Optimized database transaction pools in Flask-SQLAlchemy and added thread-safe parameter configurations. Pre-mapped numerical classifications on the client-side to

decrease data parsing load on the backend.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



LOCUST						
http://localhost:5001		RUNNING	200	80.38	0%	EDIT STOP RESET
STATISTICS	CHARTS	FAILURES	EXCEPTIONS	CURRENT RATIO	DOWNLOAD DATA	LOGS
# Failures	Method	Name	Message	First Seen	Last Seen	

Prediction History Logs

Total Logged: 1143

DATE & TIME	APPLICANT DETAILS	PROFILE	FINANCIALS	MODEL DETAILS	STATUS	ACTIONS
2026-07-05 15:45:50	sriya s@gmail.com Credit Officer	Gender: Female Education: Graduate Married: Yes	Income: \$20,000/mo Loan: \$500,000 Term: 360 Days	XGBoost Classifier Prob: 95.2%	Approved	View Delete
2026-07-05 15:44:05	Test Runner runner@test.com Applicant	Gender: Male Education: Graduate Married: Yes	Income: \$6,000/mo Loan: \$200,000 Term: 360 Days	XGBoost Classifier Prob: 90.0%	Approved	View Delete
2026-07-05 15:44:05	Test Runner runner@test.com Applicant	Gender: Male Education: Graduate Married: Yes	Income: \$6,000/mo Loan: \$200,000 Term: 360 Days	XGBoost Classifier Prob: 90.0%	Approved	View Delete
2026-07-05 15:44:05	Test Runner runner@test.com Applicant	Gender: Male Education: Graduate Married: Yes	Income: \$6,000/mo Loan: \$200,000 Term: 360 Days	XGBoost Classifier Prob: 90.0%	Approved	View Delete
2026-07-05 15:44:05	Test Runner runner@test.com Applicant	Gender: Male Education: Graduate Married: Yes	Income: \$6,000/mo Loan: \$200,000 Term: 360 Days	XGBoost Classifier Prob: 90.0%	Approved	View Delete
2026-07-05	Test Runner runner@test.com Applicant	Gender: Male Education: Graduate Married: Yes	Income: \$6,000/mo Loan: \$200,000 Term: 360 Days	XGBoost Classifier Prob: 90.0%	Approved	View Delete

Predict Applicant Creditworthiness in Seconds

Smart Lender is an advanced, machine learning-driven platform that evaluates applicant details (income, credit history, background) using optimized classification algorithms to automate and fast-track loan eligibility decisions.

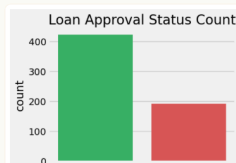
[Try Predictor Now](#)

Exploratory Data Analysis Dashboard

Univariate and bivariate analysis was performed on the loan prediction dataset to understand feature distributions and inter-variable relationships before model training.

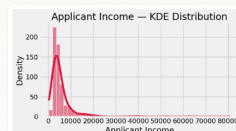
Loan Approval Distributions

Visualizing class balances for approved vs. rejected loans.



Applicant Income Distribution

KDE showing right-skewed income distribution → log-transformed.



Algorithm Performance

Comparison of trained classification models. The best-performing model is deployed for inference.

Decision Tree Classifier	Decision Tree
Train: 82.1%	Test: 77.2%
Random Forest Classifier	Random Forest
Train: 88.6%	Test: 79.7%
K-Nearest Neighbors	KNN
Train: 81.3%	Test: 74.8%

Loan Eligibility Predictor

Please fill out the multi-step form below to check loan status eligibility. All fields are parsed and standardized.

1

2

3

Step 1: User Profile

Full Name *

e.g. John Doe

Email Address *

e.g. john.doe@bank.com

User Role *

Credit Officer

Next →